End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<Btech CSAI/ECEAI/AI-ML>, Semester:<II>
(May, 2024)

Subject Code: BAS 108 | **Subject: Probability and Statistics**

Time :3 Hours | **Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) If A and B are independent and $P(A) = P(B|A) = \frac{1}{4}$, Find $P(A \cup B)$?

(b) Find the effect of change of origin and scale on moment generating function.

(c) Let X and Y have joint pdf:

$X \rightarrow$ / $Y \downarrow$	-1	0	1
0	a	2a	a
1	3a	2a	a
2	2a	a	2a

Find the condition al distributi on of X given Y = 2.

(d) Let $f(x,y) = x+y;\ 0<x<1\ ;0<y<1$. Are X and Y independent. Justify?

(e) If X has exponential distribution with mean 2, find $P(X<1|X<2)$.

(f) A random variable X has a uniform distribution over (-3,3), find $P(|X|<2)$.

(g) Explain Type I and Type II error.

(h) Differentiate between one – tailed and two – tailed test.

UNIT-I

Q2 (5+5)

(a) A random variable X has the density function $f(x) = k.\frac{1}{1+x^2}$, $-\infty < x < \infty$. Determine k and the cumulative distribution function.

(b) Urns I and II each contain two white and two black balls. One ball is selected from urn I and transferred to urn II ; then one ball is drawn from urn II and turns out to be white. What is the probability that the transferred ball was white.

Q3 (5+5)

(a) Find the mode of the following distribution by using Method of Grouping

Size	4	5	6	7	8	9	10	11	12	13
frequency	2	5	8	9	12	14	14	15	11	13

(b) Find $E[(X-1)^2]$ for the following distribution:

x	0	1	2	3
f(x)	1/3	1/2	1/24	1/8

UNIT-II

Q4 (2+3+5)

Let X and Y be jointly continuous random variables that have the joint pdf

$$f_{XY}(x,y) = \begin{cases} cx+1, & x,y \geq 0;\ x+y<1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the constant c.

(b) Find the marginal pdf $f_X(x)$.

(c) Find $P(Y < 2X^2)$.

Q5 (3+7)

A perfectly symmetric die that is equally likely to land on any of its 6

	sides is thrown twice. Let X be the upturned value of the first throw, and let Y be the sum of the two upturned values. (a) Find the joint distribution of X and Y. (b) Find $\rho(X, Y)$.	
UNIT-III		
Q6	(a) Determine the expected value of the binomial distribution by definition of expected value. (b) Prove that, if one of the regression coefficients is greater than unity, the other must be less than unity.	(5+5)
Q7	(a) Find the moment generating function of the normal distribution. (b) Obtain the rank correlation coefficient for the following data:	(5+5)

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

UNIT-IV		
Q8	(a) The following is the distribution of the daily number power failures reported in a city.	(5+5)

No. of Power Failures	0	1	2	3	4	5	6	7	8	9
No. of days	9	43	64	62	42	36	22	14	6	2

Test the goodness of fit of Poisson distribution at 5% LOS.

(b) It is found that 250 errors in the randomly selected 1000 lines of code from Team A and 300 errors in 800 lines of code from Team B. Can we assume that team B's performance is superior to that of A.

Q9	(a) A professor wants to knowif her introductory statistics class has a good grasp of basic math. Six students are chosen at random from the class and given a math proficiency test. The professor wants the class to be able to score above 70 on the test. The six students get the following scores: 62, 92, 75, 68, 83, 95. Can the professor have 90% confidence that the mean score for the class on the test would be above 70. (b) The operations manager of a company that manufactures tires wants to determine whether there are any differences in the quality of work among the three daily shifts. She randomly selects 496 tires and carefully inspects them. Each tire is either classified as perfect, satisfactory, or defective, and the shift that produced it is also recorded. The two categorical variables of interest are the shift and condition of the tire produced. The data can be summarized by the accompanying two-way table. Does the data provide sufficient evidence at the 5% significance level to infer that there are differences in quality among the three shifts?	(5+5)

	Perfect	Satisfactory	Defective
Shift 1	106	124	1
Shift 2	67	85	1
Shift 3	37	72	3

Critical values of Standard Normal Distribution: $|Z_{0.05}| = 1.96(\text{ two} - \text{tailed}), Z_{0.05} = 1.645(\text{right tailed}).$

Critical values of t-distribution: $t_{0.05} = 2.571(\text{at 5 df, two} - \text{tailed}), t_{0.05} = 2.015(\text{at 5 df, one} - \text{tailed}).$

Critical values of χ^2-distribution (5% significance level): $\chi^2_{0.05} = 11.09$ (at 5df), $\chi^2_{0.05} = 15.507$ (at 8df).